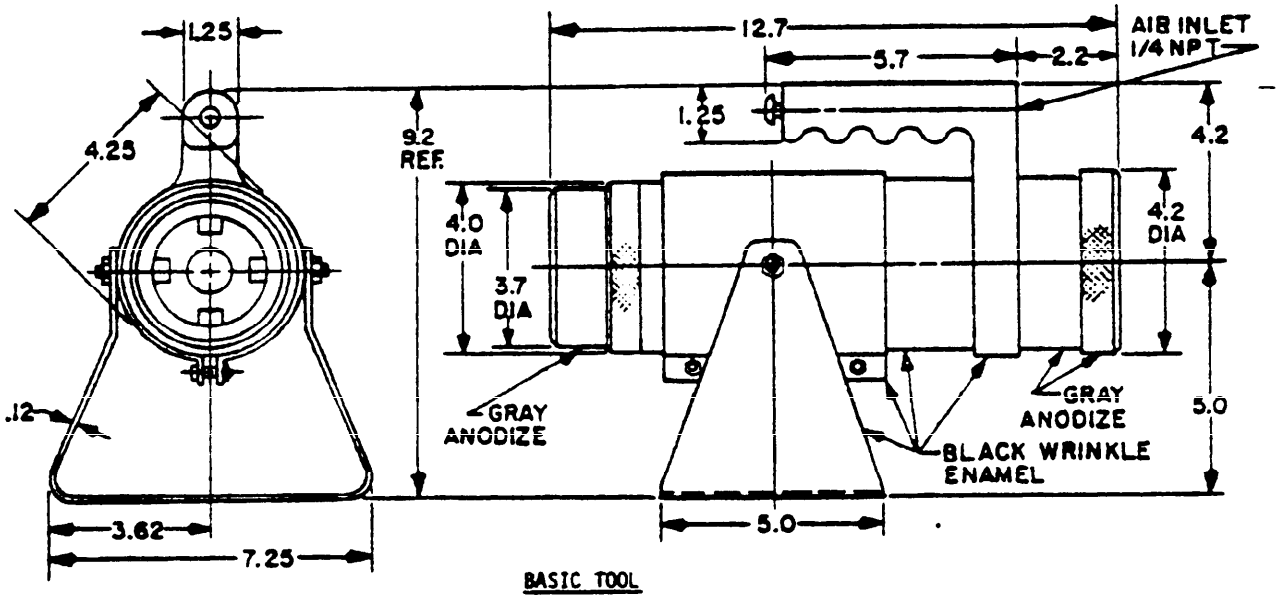
MILITARY SPECIFICATION SHEET

CRIMPING TOOLS, TERMINAL, HAND OR POWER ACTUATED, WIRE TERMINATION:

PNEUMATIC TOOL FOR WIRE BARREL SIZES 0000 THROUGH 8.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the crimping tools described herein shall consist of this document and the latest issue of specification MIL-C-22520.



BASIC TOOL

Part number
M22520/23-01

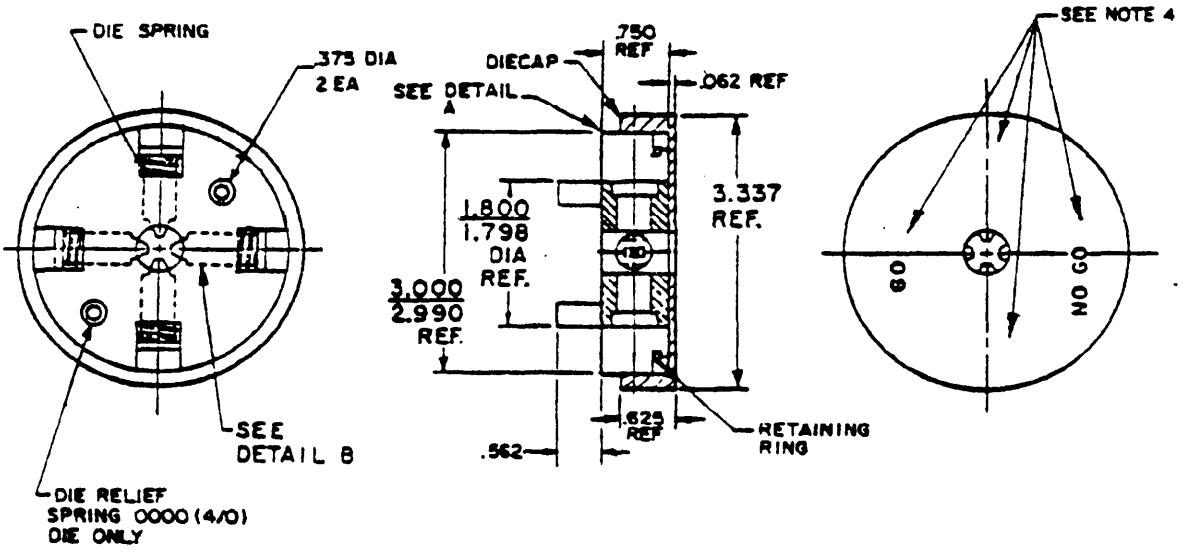
INCHES	MM	INCHES	MM
.12	3.0	4.2	107.
1.25	31.7	4.25	108.0
2.2	56.	5.0	127.
3.62	91.9	5.7	145.
3.7	94.	7.25	184.2
4.0	102.	9.2	234.
		12.7	323.

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Unless otherwise specified, tolerance is ±.125 (3.18 mm).
4. Outline configuration optional within maximum dimensions shown.
5. For applicable dies and locators see figures 2 and 3.
6. For inspection gaging of dies see Department of Commerce Commercial Standard C58.
7. Part number to be permanently and legibly stamped on body.

FIGURE 1. Pneumatic tool, for wire barrel sizes 0000 through 8.

FSC 5130

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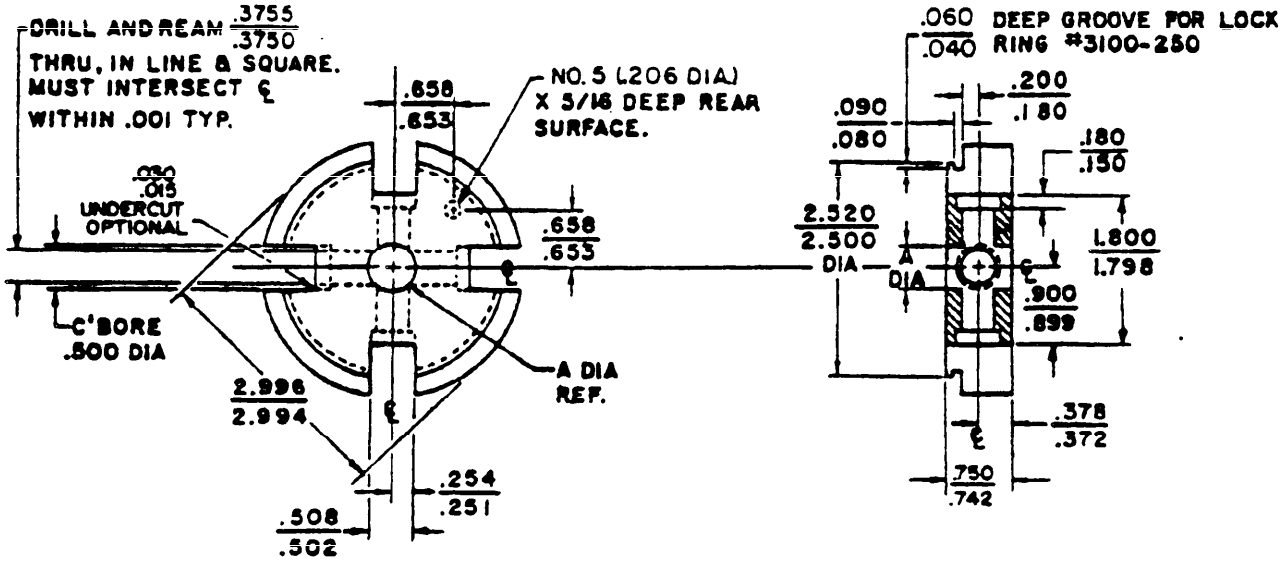


Part number	Contact size	Die gaging limits	
		Go	No Go
M22520/23-02	8	.130	.136
M22520/23-03	6	.171	.178
M22520/23-04	4	.195	.202
M22520/23-05	0	.325	.332
M22520/23-06	00	.351	.358
M22520/23-07	0000	.425	.432

INCHES	MM	INCHES	MM
.062	1.57	.358	9.09
.130	3.30	.425	10.80
.136	3.45	.432	10.97
.171	4.34	.625	15.88
.178	4.52	.750	19.05
.195	4.95	1.798	45.67
.202	5.13	1.800	45.72
.325	8.26	2.990	75.95
.332	8.43	3.000	76.20
.351	8.92	3.337	84.76

FIGURE 2. Die assembly.

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DETAIL A. Indenter housing.

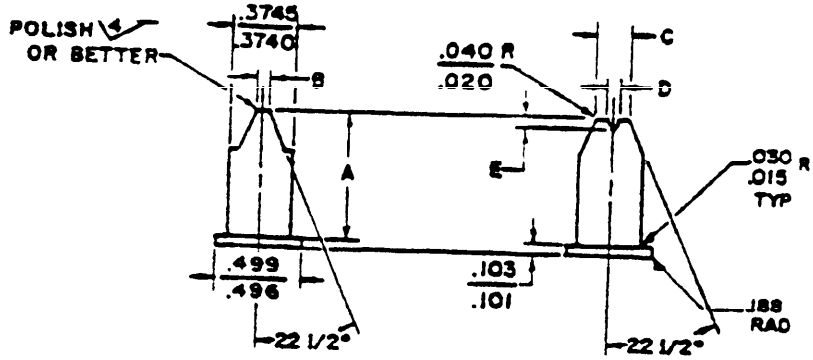
Contact size	A Dia. +.010 (.25) -.000 (.00)
8	.500 (12.70)
6	.500 (12.70)
4	.650 (16.51)
0	.690 (17.53)
00	.850 (21.59)
0000	.875 (22.22)

INCHES	MM	INCHES	MM
.001	.03	.378	9.60
.005	.13	.500	12.70
.015	.38	.502	12.75
.030	.76	.508	12.90
.040	1.02	.653	16.59
.060	1.52	.658	16.71
.080	2.03	.742	18.85
.090	2.29	.750	19.05
.150	3.81	.899	22.83
.180	4.57	.900	22.86
.200	5.08	1.798	45.67
.206	5.23	1.800	45.72
.251	6.38	2.500	63.50
.254	6.45	2.520	64.01
.372	9.45	2.994	76.05
.3750	9.525	2.996	76.10
.3755	9.538		

FIGURE 2. Die assembly - Continued.

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INCHES	MM
.015	.38
.020	.51
.030	.76
.040	1.02
.101	2.57
.103	2.62
.188	4.78
.3740	9.500
.3745	9.512
.496	12.60
.499	12.67



DETAIL B. Indenter.

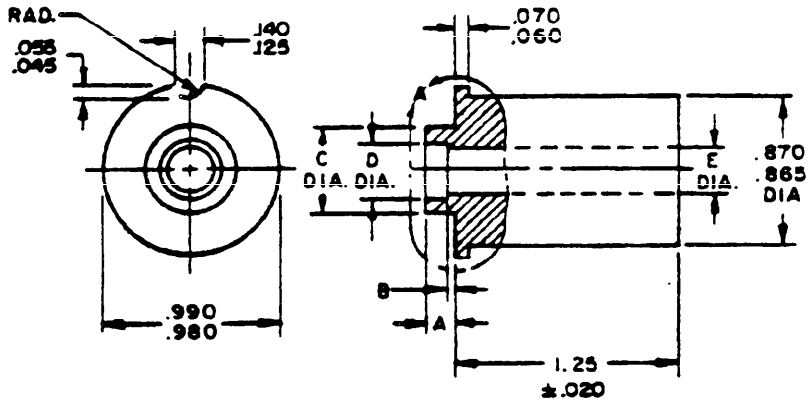
Contact size	A ±.0005 (.013)	B ±.002 (.05)	C ±.002 (.05)	D ±.002 (.05)	E +.000 -.025 (.64)
8	.833 (21.16)	.050 (1.27)	.170 (4.32)	.055 (1.40)	.055 (1.40)
6	.812 (20.62)	.060 (1.52)	.174 (4.42)	.060 (1.52)	.060 (1.52)
4	.800 (20.32)	.065 (1.65)	.187 (4.75)	.062 (1.57)	.062 (1.57)
0	.735 (18.67)	.094 (2.39)	.198 (5.03)	.070 (1.78)	.070 (1.78)
2/0	.722 (18.34)	.100 (2.54)	.210 (5.33)	.075 (1.91)	.075 (1.91)
4/0	.685 (17.40)	.130 (3.30)	.250 (6.35)	.090 (2.29)	.090 (2.29)

1/ Metric equivalents are in parentheses.

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Unless otherwise specified, tolerance is ±.005 (.13 mm).
4. Steel stamp contact size and go-no go dimension on die cap.

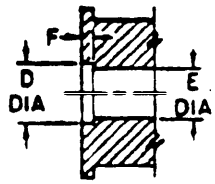
FIGURE 2. Die assembly - Continued.

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LOCATES CONTACTS

MS90453, MS90454, M39029/29, M39029/30



VIEW A

LOCATES CONTACT

MS90559-*
and
MS90560-*

INCHES	MM	INCHES	MM
.020	.51	.140	3.56
.045	1.14	.865	21.97
.055	1.40	.870	22.10
.060	1.52	.980	24.89
.070	1.78	.990	25.15
.125	3.18	1.25	31.7

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Unless otherwise specified, tolerance is ±.005 (.13 mm).
4. Steel stamp contact size and part number on locator face.

FIGURE 3. Locator.

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Part numbers and dimensions. 1/

Part number	Contact size	MS contact number	A ±.005 (.13)	B ±.005 (.13)	C Dia. ±.005 (.13)	D Dia. +.010 (.25) -.000	E Dia. +.010 (.25) -.000	F ±.005 (.13)	Wire test current (amps)	Maximum voltage drop (millivolts)
M22520/23-09	8	M39029/29-8-8 " /30-8-8 MS-90453-8-8 MS-90454-8-8	.170 (4.32)	.075 (1.91) (was .080) (2.03)	.495 (12.57)	.316 (8.03)	.266 (6.76)	---	46	5.0
M22520/23-10	6 6N 6G 6 6G	MS-90559-11 " -12 " -14 MS-90560-7 " -8	---	---	---	.348 (8.84)	.316 (8.03)	.050 (1.27)	60	4.5
M22520/23-11	4	M39029/29-4-4 " /30-4-4 MS-90453-4-4 MS-90454-4-4	.170 (4.32)	.065 (1.65)	.495 (12.57)	.420 (10.67)	.343 (8.71)	---	80	4.0
M22520/23-12	4 4N 4G 4 4G	MS-90559-8 " -9 " -13 MS-90560-5 -9	---	---	---	.423 (10.74)	.338 (8.59)	.050 (1.27)	80	4.0
M22520/23-13	1/0	M39029/29-0-0 " /30-0-0 MS-90453-0-0 MS-90454-0-0	.170 (4.32)	.025 (.64)	.675 (17.14)	.615 (15.62)	.520 (13.21)	---	125	3.0
M22520/23-14	1/0 1/0N 1/0	MS-90559-5 " -6 MS-90560-3	---	---	---	.615 (15.62)	.520 (13.21)	.050 (1.27)	125	3.0
M22520/23-15	2/0 2/0N 2/0	MS-90559-3 " -4 MS-90560-2	---	---	---	.684 (17.37)	.660 (16.76)	.050 (1.27)	185	8.0
M22520/23-16	4/0 4/0N 4/0	MS-90559-1 " -2 MS-90560-1	---	---	---	.787 (19.99)	.754 (19.15)	.050 (1.27)	225	6.0

1/ Metric equivalents are in parentheses.

FIGURE 3. Locator - Continued.

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REQUIREMENTS:

Dimensions and configuration: See figures 1 through 3.

Materials and finishes:

Basic tool:
Handle and front housing: 356 aluminum, machine finish 63, heat treat T6, black wrinkle enamel finish.
Body: 356 aluminum, machine finish exterior 63, interior 32, heat treat T6, black wrinkle enamel finish.
Cover nut: 356 aluminum, machine finish 63, heat treat T6, anodize type II finish.
Booster body: 356 aluminum, machine finish exterior 63, interior 32, heat treat T6, anodize type II finish.
Booster cap: 6061 aluminum, machine finish 63, heat treat T6, anodize type II finish.
Piston guide: 356 aluminum, machine finish 63, heat treat T6, anodize type II finish.
Piston: 6061 aluminum, machine finish 32, heat treat T6, (2 required).
Piston rods: 1010-1020 CRS, machine finish 63, heat treat-case harden C55-60, .005 (.13) - .015 (.38) deep, black oxide finish, (2 required).
Cycling valve: 303 stainless, machine finish 32, (2 required).
Retainer: 17-4 stainless, machine finish 63, heat treat 200,000-220,000 PSI, (1 required).
Arm: S-7 tool steel, machine finish 63, heat treat C54-56 Rockwell black oxide, (4 required).
Yoke: 17-4 stainless, machine finish 63, heat treat 200,000-220,000 PSI, (4 required).
Anchor: 17-4 stainless, machine finish 63, heat treat 200,000-220,000 PSI, (1 required).
Cycling valve outer and inner spring: 302 stainless, (2 required).
Trigger housing: 6061 aluminum, machine finish 63, heat treat T6, anodize type II finish.
Trigger slide: 303 stainless, machine finish 63.
Return spring: Music wire, zinc plated.

Die assembly:
Die cap: 356 aluminum, machine finish 63, heat treat T6, anodize type II finish.
Retaining ring: Industrial retaining ring number 3100-250.
Die spring: 302 stainless, 5 coils of .022 (.56) dia. wire, .385 (9.78) I.D x .562 (14.29) long, (4 required).

Indenter housing: 17-4 PH stainless, machine finish 63.

Indenters: S-7 tool steel, machine finish 63, heat treat Rockwell C54-56, (4 required).

Locators: 2024 aluminum, machine finish exterior 63, interior 32.

Bench mount: Commercial grade steel, black wrinkle enamel finish.

Air pressure required: 90-120 pounds per square inch.

Crimping requirements shall conform to 3.4.1J of MIL-C-22520 with the following exceptions:

a. The indenter closure selector need not have a positive detent at each setting.
b. A crimp joint may be obtained with aid of a turret.

Weight: Less dies and locators, 17 pounds.

Part numbers: M22520/23-01 basic tool.
M22520/23-02 thru -07 die assemblies.
M22520/23-09 thru -16 locators.

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Qualification inspection: Qualification inspection shall consist of those examinations and tests specified in MIL-C-22520 for type I tools, except that the following tests shall be modified or omitted:

a. Ratchet: The full cycle mechanism shall be tamper proof so that it cannot be disengaged prior to or during the crimp cycle. There is no ratcheting mechanism in the power unit. Full cycle mechanism operates by the use of mechanically operated valves.
b. Humidity (steady state): Omit.
c. Full cycling mechanism: Replaces ratcheting mechanism. The difference between the first and third crimp measurements shall not exceed .010 inch (.25) when tested as follows: with 90 lbf/in^2 air pressure supplied to the tool, crimp a .312 inch (7.92) diameter soft copper rod in a number 4 die; measure and record the rod thickness between opposing indents. Starting with a low air pressure that will not allow the indenters to open when crimped on the rod, slowly increase the air pressure until the tool completes its cycle and the indenters open. At this same air pressure, crimp the rod a third time; measure and record the thickness between opposing indents. The difference between the first and third measurements shall not exceed .010 inch (.25).
d. Ratcheting mechanism, type I tools: Omit.
e. High compression force: Omit.
f. Compression force: Omit.
g. Deformation and axial concentricity: After crimping to wire (wired contacts), the total indicator reading (TIR) shall not exceed 0.030 inch (.76) for sizes 0000 thru 8.
h. Voltage drop: See table in figure 3.
i. Tensile strength (crimp terminations): Shall meet the tensile strength (type AO of table X), initial conditional values, of MIL-C-39029.
j. Low temperature crimp: Omit.
k. Shock: Omit.
l. Life (type I tools): The tool shall be subjected to 1,000 crimping cycles in accordance with the following: Only one tool shall be tested. The tool shall be equipped with the largest die it is designed to accommodate (0000). Commercial leaded bronze test rods with a Rockwell hardness of B50+2 shall be advanced 0.500 inch (12.7) after each crimp. After every 250 cycles, the tool shall be operated to crimp six (0000) wire-contact assemblies. Three of the contacts shall be crimped to (0000) wire and three to (00) wire using the appropriate bushing. The tool shall require no adjustments or repairs during the 1,000 cycles, but it shall be permissable to remove foreign matter and lubricate the tool every 250 cycles. The wired contacts crimped during the life test shall meet the tensile strength requirements of table X, initial condition values, of MIL-C-39029.
m. Salt spray: Not applicable.
n. Qualification test samples: The test samples shall consist of the following:

Type I tools.

1. Two of the same part numbered basic tools with dies and locators for which qualification is desired shall be tested except only one basic tool shall be subjected to the life test for type I tools of MIL-C-22520 as outlined in "M" above; this admits qualification approval of both the basic tool with dies and locators tested.
2. After approval of items in 1., qualification approval of additional dash numbered dies and locators shall be obtained by testing two of each additional dies and locators with two previously qualified basic tools.
3. Qualification approval of dies and locators alone shall be obtained by testing two of each dash numbered dies and locators with two previously qualified basic tools.

o. Indenter and die return operation (full cycling) (type I tools): Operate the tool again to make up 6 wired assemblies for each contact size and wire plating accommodated by the tool and dies under test. Three of these shall be with the largest size wire accommodated by the contact as specified in table X of MIL-C-39029, and 3 with the smallest wire and the appropriate bushing. Upon completion of each operation of the tool and release of the applied air pressure, the tool shall meet the requirements for the indenter and die return operation in MIL-C-22520/23.
p. Marking of name plates and data plates - type I tools: Omit.
q. Gaging: Activate the tool to the fully closed position. The "GO" gage shall be freely inserted in the space between the opposing closed indenters of the dies. The "NOGO" gage shall not be insertable between the opposing indenters of the dies.

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r. Quality conformance:
   a. Group A inspection: The quality conformance inspection for group A shall be performed as listed in table III of MIL-C-22520, with the following exceptions:
      1. Handle, indentor, and die return operation (full cycling): Shall conform to MIL-C-22520/23, qualification inspection, paragraph C.
      2. Ratcheting mechanism: Omit.
      3. High compression force: Omit.
      4. Deformation of crimped connection: Shall conform to MIL-C-22520/23, qualification inspection, paragraph g.
      5. Axial concentricity: Shall conform to MIL-C-22520/23, qualification inspection, paragraph g.
      6. Tensile strength: Shall conform to MIL-C-22520/23, qualification inspection, paragraph i.
   b. Group B inspection: The quality conformance inspection for group B shall be performed as listed in table IV of MIL-C-22520, with the following exceptions:
      1. Compression force: Omit.
      2. Deformation of crimp connection: Shall conform to MIL-C-22520/23, qualification inspection, paragraph g.
      3. Axial concentricity: Shall conform to MIL-C-22520/23, qualification inspection, paragraph g.
      4. Voltage drop: Omit.
      5. Tensile strength (1st): Shall conform to MIL-C-22520/23, qualification inspection, paragraph i.
      6. Low temperature crimp: Omit.
      7. Tensile strength (2nd): Omit.
      8. Shock: Omit.
      9. Handle, indenter and die return operation (full cycling): Shall conform to MIL-C-22520/23, qualification inspection, paragraph c.

NOTE: Pressure required to crimp a 4/0 contact with appropriate bushing to a 2/0 wire may require 100 pounds per square inch.

Custodians:
Army - ER
Navy - AS
Air Force - 85

Review activities:
Army - AR, AV, GL, MI, WC
Navy - EC, OS, SH
Air Force - 11, 12, 17, 99
DLA - ES

User activities:
Army - AT, ME
Navy - MC, YD
Air Force - 19

Preparing activity:
Navy - AS

Agent:
DLA - ES

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